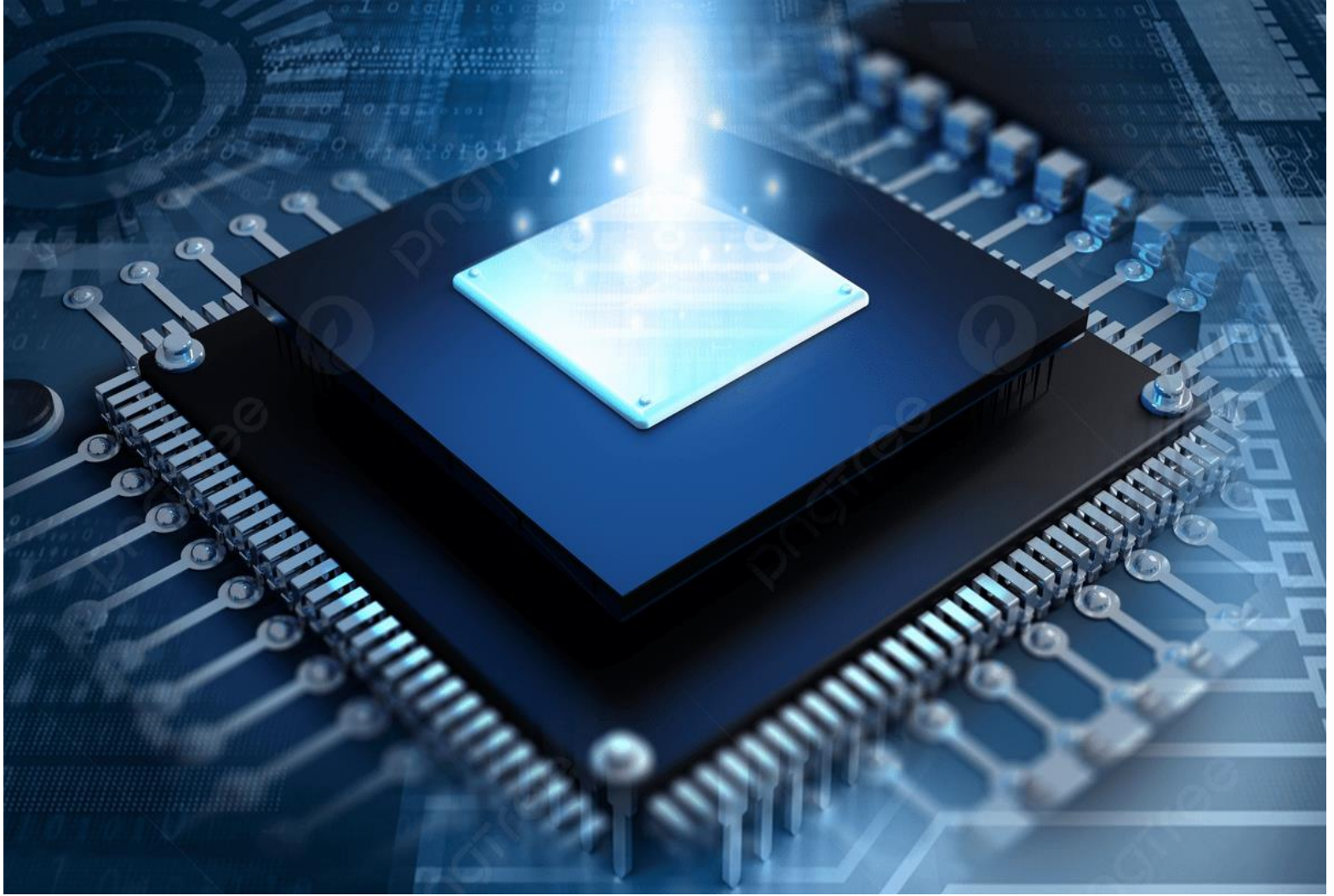




Advanced UVM-Inspired Class-Based Verification Environment for ALU using SV

GitHub

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Abstract

This project presents the design and implementation of an advanced class-based verification environment for a pipelined Arithmetic Logic Shift Unit (ALU) using System Verilog and Questa Sim.

The verification environment was developed following a UVM-inspired object-oriented methodology to thoroughly validate the functional correctness and robustness of the ALU under constrained-random stimulus.

The ALU supports a wide range of operations including arithmetic (addition, multiplication), logical (OR, XOR), shift/rotate, reduction, bypass modes, and invalid opcode handling with proper LED behavior. To achieve comprehensive verification, a complete reusable environment was built consisting of Transaction, Generator, Driver, Monitor, Scoreboard, Coverage, Environment, and System Verilog Assertions (SVA) components.

A self-checking scoreboard with a golden reference model was implemented to automatically compare expected and actual outputs. Latency-aware synchronization mechanisms were developed to correctly align transactions with the DUT's 2-cycle pipeline delay. Constrained-random stimulus, functional coverage with cross-coverage, code coverage, and assertion-based verification were employed to ensure verification closure.

The project successfully achieved 100% code coverage, functional coverage, and assertion coverage while overcoming several challenges related to virtual interface binding, transaction synchronization, and coverage closure. This work demonstrates advanced verification concepts including constrained-random testing, coverage-driven verification, scoreboard design, and reusable object-oriented testbench architecture commonly used in modern ASIC verification flows

Project Overview

The verification of digital designs has become a critical stage in modern ASIC and FPGA development cycles due to the ever-increasing complexity of hardware systems and the high cost of functional failures post-fabrication. This project focuses on the development of a complete **class-based verification environment** for a pipelined Arithmetic Logic Shift Unit (ALU) using System Verilog and Questa Sim.

The primary objective is to thoroughly verify the functional correctness, reliability, and robustness of the ALU under both directed and constrained-random test scenarios. The ALU supports a rich set of operations including arithmetic (addition, multiplication), logical (OR, XOR), shift/rotate, reduction, bypass modes, Asynchronous reset handling, and invalid opcode detection with LED toggling.

To achieve comprehensive verification closure, a reusable **UVM-inspired** verification environment was implemented using advanced object-oriented System Verilog concepts. The environment consists of several key components: Transaction, Generator, Driver, Monitor, Scoreboard, Coverage Collector, Environment, and System Verilog Assertions (SVA).

Transaction-level communication between components is achieved using mailboxes and virtual interfaces. The verification flow starts with constrained-random stimulus generation in the Generator, which is then driven to DUT through the Driver. The Monitor observes the DUT responses after the 2-cycle pipeline latency and forwards them to both the Scoreboard and Coverage components. A self-checking Scoreboard with a golden reference model ensures automatic comparison between expected and actual outputs, while advanced functional coverage models with cross-coverage guarantee that all operational modes and corner cases are exercised.

This project demonstrates industry-standard verification practices including constrained-random testing, coverage-driven verification, assertion-based verification, scoreboard design, and reusable testbench architecture.

ALU Design & Features

Design Under Test (DUT)

The Design Under Test (DUT) in this project is a pipelined **Arithmetic Logic Shift Unit (ALU)** implemented in System Verilog RTL. The ALU is designed to perform a wide range of operations including arithmetic, logical, shift, rotate, reduction, and bypass functions, along with proper handling of reset and invalid operations.

The ALU operates synchronously with the system clock and features a **2-cycle pipeline latency** — inputs are sampled on one clock cycle, and the result is registered and available on the following cycle. This pipelined behavior added an important verification challenge regarding transaction synchronization between the Driver and Monitor.

Supported Operations

- **Arithmetic Operations:** Addition (ADD), Multiplication (MULT)
- **Logical Operations:** Bitwise OR, Bitwise XOR
- **Shift & Rotate Operations:** Shift Left/Right and Rotate Left/Right with programmable direction
- **Reduction Operations:** Reduction OR and Reduction XOR on operand A or B
- **Bypass Modes:** Direct bypass for input A or input B (with priority)
- **Special Conditions:** Asynchronous reset handling and invalid opcode detection with LED indication

Input Signals

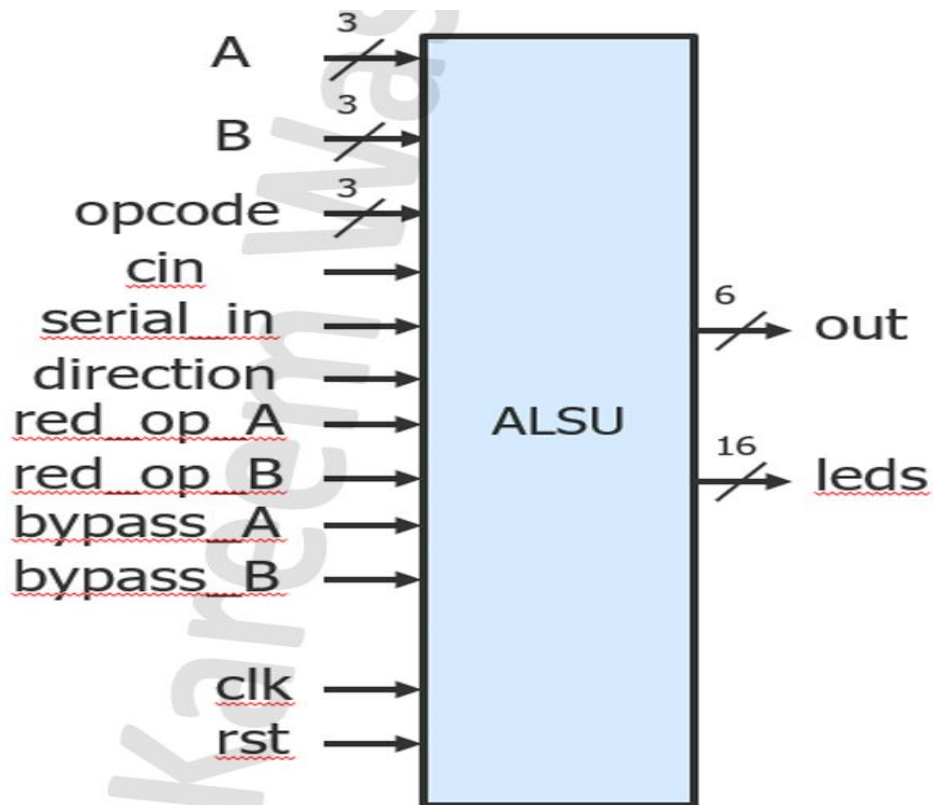
- clk: System clock
- rst: Active-high synchronous reset
- A, B: Signed 3-bit input operands
- Opcode: 3-bit operation selection
- serial_in: Serial input for shift/rotate operations
- direction: Direction control for shift and rotate
- red_op_A: Enable reduction operation on operand A
- red_op_B: Enable reduction operation on operand B
- bypass_A: Enable bypass for operand A (highest priority)
- bypass_B: Enable bypass for operand B

Output Signals

- out: 6-bit ALU result
- leds: Status LEDs (toggles on invalid operations)

The ALU was intentionally designed with multiple operational modes, priority logic (bypass > reduction > normal), and corner-case behaviors to facilitate the application of advanced verification techniques such as constrained-random testing, cross-coverage, assertion-based verification, and pipeline-aware synchronization.

DUT Block Diagram



Verification Plan

The verification plan was developed to ensure complete coverage of all ALU functionalities, control signals, corner cases, and invalid conditions using constrained-random stimulus, self-checking mechanisms, assertions, and coverage-driven verification.

The following table summarizes the verification strategy implemented in the project:

	Feature / Functionality	Stimulus / Test Scenario	Expected Check	Assertion Verification	Functional Coverage
1	Asynchronous Reset	Toggle rst from 0 → 1	out = 0 and leds = 0	Final assertion	Reset coverage
2	Addition Operation	opcode = ADD	out = A + B	Concurrent assertion	Opcode coverage for ADD
3	Addition Overflow	Maximum signed values for A and B	Correct overflow behavior	Concurrent assertion	Cross coverage (A, B, ADD)
4	OR Operation	opcode = OR	out = A B	Concurrent assertion	Opcode coverage for OR
5	Reduction OR (A/B)	opcode = OR + red_op_A or red_op_B	Reduction result on A or B	Concurrent assertion	Cross coverage (opcode × red_op)
6	XOR Operation	Opcode = XOR	out = A ^ B	Concurrent assertion	Opcode coverage for XOR
7	Reduction XOR (A/B)	Opcode = XOR + red_op_A or red_op_B	Reduction XOR on A or B	Concurrent assertion	Cross coverage (opcode × red_op)
8	Multiplication	opcode = MULT	out = A * B	Concurrent assertion	Opcode coverage for MULT
9	Multiplication Overflow	Maximum values for A and B	Correct overflow behavior	Concurrent assertion	Cross coverage (A, B, MULT)
10	Shift Operations	opcode = SHIFT + direction control	Correct shift left/right	Concurrent assertion	Cross coverage (opcode × direction)
11	Rotate Operations	opcode = ROTATE + direction control	Correct rotate left/right	Concurrent assertion	Cross coverage (opcode × direction)
12	Bypass A / Bypass B	bypass_A = 1 or bypass_B = 1	out = A or out = B	Concurrent assertion	Bypass coverage

	Feature / Functionality	Stimulus / Test Scenario	Expected Check	Assertion Verification	Functional Coverage
13	Invalid Operations	Invalid opcodes or illegal reduction combinations	out = 0 + LED toggle	Concurrent assertion	Illegal bin coverage

Verification Methodology

The verification environment was built using a **constrained-random class-based methodology**. Randomized transactions were generated by the Generator, driven to the DUT through the Driver, and observed by the Monitor after the 2-cycle pipeline latency.

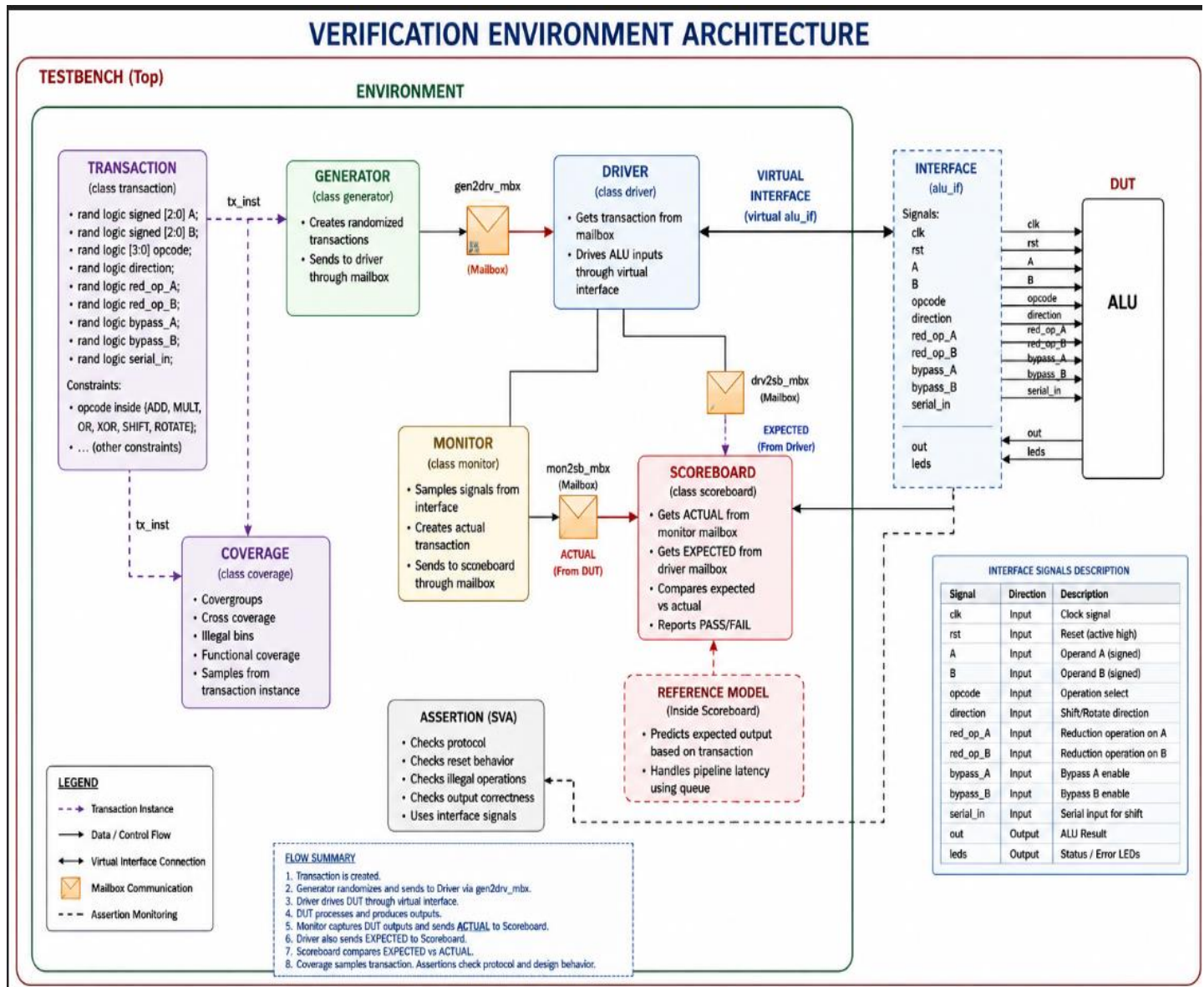
A self-checking Scoreboard with a golden reference model was used for automatic result verification. System Verilog Assertions (SVA) were implemented to check protocol correctness, reset behavior, and invalid conditions in real-time. Functional coverage with cross-coverage was employed to measure verification quality and ensure all operational modes and corner cases were exercised.

The verification process focused on:

- Arithmetic and logical correctness
- Shift, rotation, and reduction functionality
- Bypass priority handling
- Overflow conditions
- Invalid operation detection
- Pipeline synchronization
- Reset behavior
- Full coverage closure

Verification Environment Architecture

Architecture Block Diagram



Environment Components (Explanation & RTL)

1. Transaction Class:

The Transaction class serves as the main data container for the verification environment. It holds all input and output signals with intelligent randomization constraints to generate meaningful test scenarios. This class is the fundamental building block that flows through all other components.

```
Transaction.svh
1  // Importing ALL Packages
2
3  import ALU_PACKAGE::*;
4
5
6  // deifing Classes for Rndomization Process
7
8  class static Transaction;
9
10 // ===== Inputs (Randomized) =====
11
12     rand bit          rst;
13     rand logic signed [2:0] A, B;
14     rand op           opcode;
15     rand bit          serial_in;
16     rand bit          direction;
17     rand bit          red_op_A;
18     rand bit          red_op_B;
19     rand bit          bypass_A;
20     rand bit          bypass_B;
21
22
23 // ===== For Observing the output Signals =====
24
25
26     logic signed [5:0] out;
27     logic          [15:0] leds;
28
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35 // Internal Signals Will be used in Transaction Class
36
37     int          max_pos = 3, max_neg = -4, zero = 0;
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```

3. Driver Class

The Driver receives transactions from the Generator and drives them to the DUT through a virtual interface using the TEST modport. It converts high-level transactions into pin-level activity at the negative edge of the clock for proper synchronization.

```
Driver.svh
1  /////////////// Importing ALL Packages
2
3  import ALU_PACKAGE::*;
4
5  class Driver;
6
7  /////// Connecting the driver with the DUT in the TB modport(TEST)
8
9  virtual ALU_IF vif;
10
11  ///// Mail Box for taking the tr_driv_rand from generator
12
13  mailbox GEN2DRV;
14
15  ///// Mail Box for taking the tr_driv_rand to Score board (Actual Data)
16
17  mailbox DRV2SCB;
18
19  /// Consructor
20
21  function new(mailbox GEN2DRV, virtual ALU_IF vif, mailbox DRV2SCB);
22
23      this.GEN2DRV = GEN2DRV;  ///// for passing this mail to the class one
24      this.vif = vif;
25      this.DRV2SCB = DRV2SCB;
26
27  endfunction
28
29
30  /////////////// Task for Driving the Randomized tr_drivansction to the Dut
31
32  task run();
33
34      forever
35          begin
36              Transaction tr_driv = new();  ///// for putting the re_rand on it
37              GEN2DRV.get(tr_driv);
38
39              /////////////// Passing the Rand_tr_driv to the Dut
40
41              @(negedge vif.clk)
42
43              vif.rst      <= tr_driv.rst;
44              vif.A        <= tr_driv.A;
45              vif.B        <= tr_driv.B;
46              vif.serial_in <= tr_driv.serial_in;
47              vif.opcode    <= tr_driv.opcode;
48              vif.direction <= tr_driv.direction;
49              vif.red_op_A  <= tr_driv.red_op_A;
50              vif.red_op_B  <= tr_driv.red_op_B;
51              vif.bypass_A  <= tr_driv.bypass_A;
52              vif.bypass_B  <= tr_driv.bypass_B;
53
54              /////////////// FOR sending the Data to the Score board to compare it with Monitor
55              DRV2SCB.put(tr_driv);
56
57          end
58      endtask
59
60 endclass
```

4. Monitor Class

The Monitor passively observes the DUT outputs using the MONITOR modport of the virtual interface. It waits for the 2-cycle pipeline latency, samples the response, and forwards the actual transaction to both the Scoreboard and Coverage components.

```
Monitor.svh
1  // Importing ALL Packages
2
3  import ALU_PACKAGE::*;
4
5  // Monitor Class
6
7  class Monitor;
8
9  // Connecting the Monitor with the DUT in the Monitor modport
10
11  virtual ALU_IF vif;
12
13  // Mail Box To send the Transaction Data From Monitor to Score Board
14
15  mailbox MON2SCB;
16
17  // Defining an Coverage Object to sample the Data
18
19  Coverage cov;
20
21
22  // Constructor
23
24  function new(mailbox MON2SCB, virtual ALU_IF vif);
25
26      this.MON2SCB = MON2SCB; // for passing this mail to the class one
27      this.vif = vif;
28      this.cov = new();
29
30  endfunction
31
32  // Run Task for Sampling and sending the Data from DUT to SCB and Cov
33
34  task run();
35
36      forever
37          begin
38
39              Transaction tr_mon = new();
40
41              // repeat(2)
42              @(negedge vif.clk) ; // Waiting 2 cycles one cause in rtl we sample the input
43              // Sampling the Inputs
44
45              tr_mon.rst      = vif.rst;
46              tr_mon.serial_in = vif.serial_in;
47              tr_mon.opcode   = vif.opcode;
48              tr_mon.direction = vif.direction;
49              tr_mon.red_op_A  = vif.red_op_A;
50              tr_mon.red_op_B  = vif.red_op_B;
51              tr_mon.bypass_A  = vif.bypass_A;
52              tr_mon.bypass_B  = vif.bypass_B;
53              tr_mon.A         = vif.A;
54              tr_mon.B         = vif.B;
55
56              // Sampling the Outputs
57
58              @(posedge vif.clk) ;
59              tr_mon.out       = vif.out;
60              tr_mon.leds      = vif.leds;
61
62              // Passing tr_m to score board
63
64              MON2SCB.put(tr_mon);
65          end
66      end
67  endtask
68  endclass
```

5. Scoreboard Class

The Scoreboard is the self-checking component of the environment. It receives expected transactions from the Driver and actual transactions from the Monitor, then compares them using a golden reference model to determine pass/fail status.

```
////////// Importing ALL Packages
import ALU_PACKAGE::*;

class Score_Board;

    /// Mail Box To recieve the Transaction Data From Monitor
    mailbox MON2SCB;

    /// Mail Box To recieve the Transaction Data From Driver
    mailbox DRV2SCB;

    /// Constructor
    function new(mailbox MON2SCB, mailbox DRV2SCB);
        this.MON2SCB = MON2SCB;    /// for passing this mail to the class one
        this.DRV2SCB = DRV2SCB;
    endfunction

    /// Defing the Error and Pass Counts
    int Pass_Count = 0;
    int Error_Count = 0;
    int i = 0;
    logic signed [5:0]  expec_outt;
    bit outt;

    /////////// Defining the Queues for SYNC the Mon with Drv
    Transaction  tarns_q[$] ;
    parameter  Pipe_line_stages = 2;

    // Tran_expec_queue.out ;
    Transaction tr_expected = new();
    Transaction TR_QUEUE = new();
    Transaction tr_actual = new();

    /////////// Run Task for Comparing the Expected Data from the Driver and the one From Monitor the Actual one
    task run();
        for (int r = 0; r < Pipe_line_stages ; r++)
            begin
                Transaction tr_init = new();
                tarns_q.push_back(tr_init);
            end
        forever
            begin
                /// put the Data comes from the driver on it
                DRV2SCB.get(tr_expected);
                MON2SCB.get(tr_actual);    /////////// Getting the ACTUAL FROM the Monitor

                tarns_q.push_back(tr_expected);    /////////// storing the Expec Values

                if(tarns_q.size() > 2);    /////////// Waiting 2 Cycles for SYNC
                begin
                    ..... TR QUEUE = tarns q.pop front();
                    Compare(outt,expec_outt,tr_actual,TR_QUEUE);
                    i = i + 1;

                    $display("-----");
                    $display(" iteration number = %0d",i);
                    $display("-----");

                    $display("-----");
                    $display("expec_out = %0d & actual_out = %0d at time ",expec_outt,tr_actual.out,$time);
                    $display("-----");

                    if(outt)
                        Pass_Count = Pass_Count + 1;
                    else
                        Error_Count = Error_Count + 1;
                end
            end
        end

        $display("-----");
        $display(" ***** total no of Iteration = %0d *****",i+1);
        $display("-----");
    endtask
endclass
```

6. Coverage Class

The Coverage class implements functional coverage using cover groups and cross-coverage. It collects coverage data from the Monitor to ensure all operational modes, corner cases, and combinations (such as overflow and reduction) are properly exercised.

```
////////// Importing ALL Packages

import ALU_PACKAGE::*;

////////// Class for Coverage Process

class Coverage ;

    //////////// Taking an INSTANCE from the Transaction Class to cover it
    Transaction ALU_tr_cov;

    ////////// Defining The Cover Group

    covering ALL Packages

    ALU_PACKAGE::*;

    /// Class for Coverage Process

    e ;

    //////////// Taking an INSTANCE from the Transaction Class to cover it
    Transaction ALU_tr_cov;

    ////////// Defining The Cover Group

    covergroup ALU_Coverage ;

        A_CP: coverpoint ALU_tr_cov.A iff(!ALU_tr_cov.rst) {

            bins A_Data_0      = {0};
            bins A_Data_Max    = {ALU_tr_cov.max_pos};
            bins A_Data_Min    = {ALU_tr_cov.max_neg};
            // bins Walking_Ones[] = {3'b001,3'b010,3'sb100} iff(ALU_tr_cov.red_op_A)
            bins others        = {[ALU_tr_cov.max_neg + 1 : ALU_tr_cov.max_pos - 1]};

        }

        B_CP: coverpoint ALU_tr_cov.B iff(!ALU_tr_cov.rst) {

            bins B_Data_0      = {0};
            bins B_Data_Max    = {ALU_tr_cov.max_pos};
            bins B_Data_Min    = {ALU_tr_cov.max_neg};
            // bins Walking_Ones[] = {3'b001,3'b010,3'sb100} iff(ALU_tr_cov.red_op_B)
            bins others        = {[ALU_tr_cov.max_neg + 1 : ALU_tr_cov.max_pos - 1]};

        }

        REDUCTION_A: cross OP_CP,RED_OP_A {
            option.cross_auto_bin_max = 0 ;
            bins OR_RED_OP_A      = binsof (OP_CP.OR ) && binsof (RED_OP_A[1] );
            bins XOR_RED_OP_A     = binsof (OP_CP.XOR ) && binsof (RED_OP_A[1] );
        }

        REDUCTION_B: cross OP_CP,RED_OP_B {
            option.cross_auto_bin_max = 0 ;
            bins OR_RED_OP_B      = binsof (OP_CP.OR ) && binsof (RED_OP_B[1] );
            bins XOR_RED_OP_B     = binsof (OP_CP.XOR ) && binsof (RED_OP_B[1] );
        }

        Multi_OVERFLOW: cross OP_CP,A_CP,B_CP {
            option.cross_auto_bin_max = 0 ;
            // bins Positive_over_flow = binsof (OP_CP.MULT) && binsof (A_CP.A_Data_Max ) && binsof (B_CP.B_Data_Max )
            bins Negative_over_flow = binsof (OP_CP.MULT ) && binsof (A_CP.A_Data_Min ) && binsof (B_CP.B_Data_Min );
        }

        SHIFT_LEFT_RIGHT: cross OP_CP,Direction {
            option.cross_auto_bin_max = 0 ;
            bins SHIFT_RIGHT      = binsof (OP_CP.SHIFT ) && binsof (Direction[0]);
            bins SHIFT_LEFT       = binsof (OP_CP.SHIFT ) && binsof (Direction[1]);
        }

    endcovergroup

endclass
```

7. Environment Class

The Environment acts as the top-level manager of verification architecture. It instantiates all components (Generator, Driver, Monitor, Scoreboard, Coverage), connects them using mailboxes, and controls the overall execution flow.

```
////////// Importing ALL Packages

import ALU_PACKAGE::*;

class Environment;

    // Components
    Generator      GEN;
    Driver          DRV;
    Monitor        MON;
    Score_Board    SCB;

    // Mailboxes
    mailbox GEN2DRV;
    mailbox DRV2SCB;
    mailbox MON2SCB;

    // Constructor
    // function new(virtual ALU_IF.TEST vif_T, virtual ALU_IF.Monitor vif_M);
    function new(virtual ALU_IF vif);
        // this.vif = vif;
        GEN2DRV = new();
        DRV2SCB = new();
        MON2SCB = new();

        GEN      = new(GEN2DRV);
        DRV      = new(GEN2DRV, vif, DRV2SCB);
        MON      = new(MON2SCB, vif);
        SCB      = new(MON2SCB, DRV2SCB);

    endfunction

    task run();

        fork

            GEN.run();
            DRV.run();
            MON.run();
            SCB.run();

        join_any

    endtask

endclass
```

8. Interface (ALU_IF)

The Interface defines all signals between the testbench and the DUT. It includes clocking blocks and mod ports (TEST, MONITOR, DUT) to provide clean, race-free access and proper direction control for Driver and Monitor.

```
interface ALU_IF(clk);

    input bit clk;

    ////////////////////////////////////////////////// Importing ALL Packages

    import ALU_PACKAGE::*;

    /////////////////////////////////// defining Signals ///////////////////////////////////

    /////////////// Input Signals

    bit rst;
    logic signed [2:0] A;
    logic signed [2:0] B;
    op opcode;
    bit serial_in;
    bit direction;
    bit red_op_A;
    bit red_op_B;
    bit bypass_A;
    bit bypass_B;

    /////////////// Output Signals

    logic signed [5:0] out;
    logic [15:0] leds;
```


9. System Verilog Assertions (SVA)

The SVA module contains concurrent assertions and immediate checks to verify protocol correctness, reset behavior, operation validity, and output consistency in real-time during simulation.

```
////////// Importing ALL Packages
import ALU_PACKAGE::*;

////////// REquied for testing the protocols and timing info and the corner cases in the design

module ALU_SVA (ALU_IF SVA_IF);

    ////////// Writing the Properties for the Concarnt Assertions

    ///// TESTING ALU_Main Operations

    ////////// ADDITION

    property ADDER;
        @(posedge SVA_IF.clk) disable iff(SVA_IF.rst || (SVA_IF.bypass_A) || (SVA_IF.bypass_B))

        (((SVA_IF.opcode == ADD ) && (!SVA_IF.red_op_A) && (!SVA_IF.red_op_B)))
        |-> ##2 ((SVA_IF.out == $past(SVA_IF.A,2) + $past(SVA_IF.B,2))
    endproperty

    ////////// Multiplication

    property MULTER;
        @(posedge SVA_IF.clk) disable iff(SVA_IF.rst)

        ((!SVA_IF.bypass_A) && (!SVA_IF.bypass_B) && (SVA_IF.opcode == MULT ) && (!SVA_IF.red_op_A) && (!SVA_IF.red_op_B))
        |-> ##2 (SVA_IF.out == $past(SVA_IF.A,2) * $past(SVA_IF.B,2)
        &&(SVA_IF.leds == 0));
    endproperty

    ////////// Writing the Assert Properties

    ADDITION: assert property (ADDER)
        else $error (" Testing the Addition has Failed") ;

    MULTIPLICATION: assert property (MULTER)
        else $error (" Testing the MULTIPLICATION has Failed") ;

    ORR: assert property (ORER)
        else $error (" Testing the ORING has Failed") ;

    REDUCTION_OR_A: assert property (RED_OR_AER)
        else $error (" Testing the REDUCTION_OR_A has Failed") ;

    REDUCTION_OR_B: assert property (RED_OR_BER)
        else $error (" Testing the REDUCTION_OR_B has Failed") ;

    always_comb
    begin
        if (SVA_IF.rst)
        begin
            Reset_for_Out: assert final (SVA_IF.out == 0)
                else $error (" Testing the ASYNC Reset for OUT has Failed") ;

            Reset_for_LEDS: assert final (SVA_IF.leds == 0)
                else $error (" Testing the ASYNC Reset for LEDES has Failed") ;
        end
    end
end
```

10. Testbench & Top Module

the Testbench and Top modules provide the simulation harness. They handle clock generation, DUT and Interface instantiation, Environment creation, and SVA binding to run the complete verification environment.

```

1  module ALU_TB;
2
3      import ALU_PACKAGE::*;
4
5      // Clock Generation
6      bit clk;
7
8      // Instantiate Interface
9      ALU_IF vif (.clk(clk));
10
11
12  // Instantiate DUT
13  ALU DUT (
14      .clk      (clk),
15      .rst      (vif.rst),
16      .A        (vif.A),
17      .B        (vif.B),
18      .opcode   (vif.opcode),
19      .serial_in (vif.serial_in),
20      .direction (vif.direction),
21      .red_op_A  (vif.red_op_A),
22      .red_op_B  (vif.red_op_B),
23      .bypass_A  (vif.bypass_A),
24      .bypass_B  (vif.bypass_B),
25      .out       (vif.out),
26      .leds      (vif.leds)
27  );
28
29  // Bind SVA
30
31  ALU_SVA SVA (vif);
32
33
34  parameter clk_period = 10 ;

```

```

12 // Instantiate DUT
39 end
40
41 ////////////////////////////////////////////////// INST FROM ENV
42
43 ////////////////////////////////////////////////// initial Block
44
45 initial begin
46     vif.rst = 1;
47     #(clk_period) vif.rst = 0;
48 end
49
50 initial
51     begin
52
53         $dumpvars ;
54         $dumpfile("ALU.vcd");
55
56         ENV.run();
57
58
59         #500000; // Run long enough
60
61         ENV.SCB.Score_Board_Report();
62         $stop;
63     end
64
65 endmodule
66

```

Debugging Challenges & Solutions

During the development of the verification environment, several synchronization, coverage, and functional issues were encountered. These challenges were systematically analyzed and resolved to achieve accurate verification, stable simulation, and high coverage.

The following table summarizes the major issues faced and their corresponding solutions:

Challenge	Root Cause	Solution Implemented
Expected vs Actual Output Mismatch	2-cycle pipeline latency in DUT while Scoreboard compared immediately	Implemented queue-based synchronization in Scoreboard to align expected and actual transactions
Incorrect First Few Transactions	Comparisons started before DUT pipeline filled	Added dummy transactions in Scoreboard equal to pipeline latency
Scoreboard always showing failures	Misalignment between Driver and Monitor timing	Added latency-aware delay in Driver before sending to Scoreboard
Multiple Coverage Instances	Coverage objects are created more than once	Ensured single instance of Coverage class and disabled redundant TYPE coverage
Illegal Bin Hits	Invalid opcodes generated during random testing	Refined constraints and separated valid/invalid test scenarios
Low Functional Coverage	Insufficient corner cases (overflow, reduction)	Added directed constraints and targeted cross-coverage bins
Assertion Failures during Reset	Assertions active during reset phase	Added disable iff (rst) in all concurrent assertions
Race Conditions between Driver & DUT	Driving signals too close to clock edge	Used clocking blocks and drove inputs at negege of clock
Incorrect Shift/Rotate Results	Wrong bit concatenation in golden model	Corrected shift/rotate logic in Scoreboard golden reference model

Simulation Results

The verification environment was simulated using **Questa Sim** with constrained-random stimulus generation and self-checking mechanisms. Multiple test scenarios were executed to validate all ALU functionalities, including arithmetic, logical, shift, rotate, reduction, bypass, overflow, and invalid operations.

The simulation successfully verified the DUT through seamless interaction between the Generator, Driver, Monitor, Scoreboard, Coverage, and Assertion components. All transactions were automatically checked against the golden reference model in the Scoreboard.

Simulation Flow

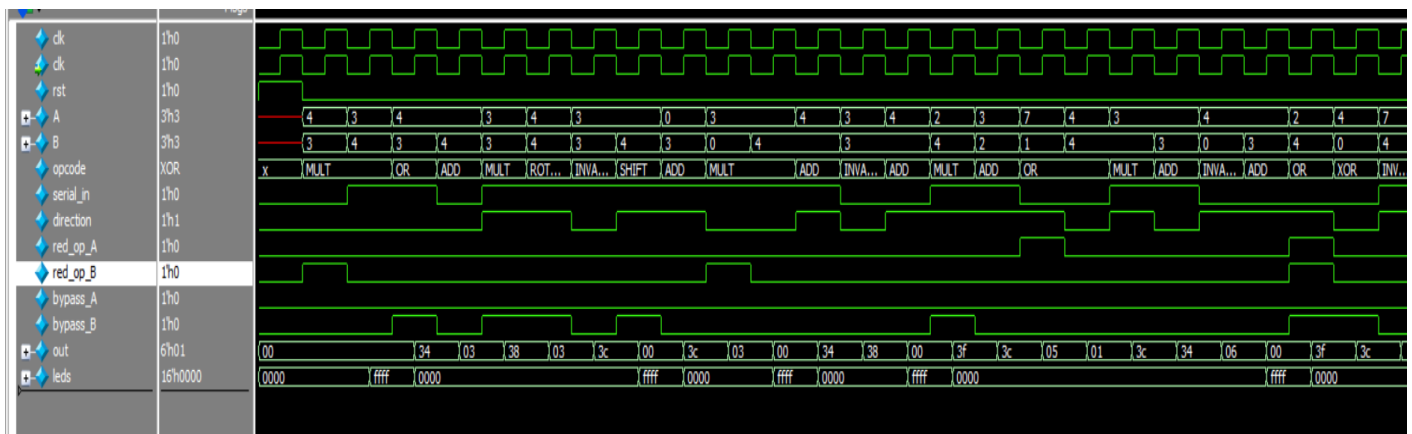
1. Constrained-random transactions were generated by the Generator.
2. Transactions were sent to the Driver via mailboxes.
3. The Driver applied inputs to the DUT using the virtual interface.
4. The Monitor sampled DUT outputs after the 2-cycle pipeline latency.
5. Expected and actual transactions were synchronized using queue buffering.
6. The Scoreboard performed automated comparison using the golden model.
7. Functional coverage and assertions were continuously collected.

Verification Metric	Result
Total Test Iterations	5000+
Passed Transactions	All valid operations passed
Failed Transactions	0
Code Coverage	100%
Functional Coverage	100%
Assertion Coverage	100%
Overflow Scenarios	Verified
Reduction Operations	Verified
Shift/Rotate Operations	Verified
Invalid Operations	Verified
Reset Behavior	Verified

Snippets from Transcript

```
# iteration number = 4998
# *****
# expect_out = 5 & actual_out = 5 at time 49985
# *****
# iteration number = 4999
# *****
# expect_out = 1 & actual_out = 1 at time 49995
# *****
# iteration number = 5000
# *****
# expect_out = -1 & actual_out = -1 at time 50005
# *****
** Error: (vsim-8565) Illegal state bin was hit at value=INVALID_6. The bin counter for the illegal bin '\ALU_PACKAGE::Coverage:ALU_Coverage .OP_CP.others' is 798.
Time: 50015 ns Iteration: 1 Region: /ALU_PACKAGE::Environment::run
# *****
# The Number OF Correct Operations =5000 at time 500000
# *****
# The Number OF Failed Operations =0 at time 500000
# *****
```

Random Snippets from Wave forms



Coverage Results

Code Coverage

File: testbench_top_top
Coverage Report by DU with details

```
=====
=== Design Unit: work.ALU
=====
Branch Coverage:
  Enabled Coverage      Bins    Hits    Misses  Coverage
  -----
  Branches              25      25      0      100.00%
```

Branch Coverage for Design Unit work.ALU

Line	Item	Count	Source
File ALU.sv			
-----IF Branch-----			
32		1005	Count coming in to IF
32	1	2	if (rst) begin
44	1	1003	end else begin
Branch totals: 2 hits of 2 branches = 100.00%			
-----IF Branch-----			
62		1002	Count coming in to IF
62	1	48	if (bypass_A_reg) begin
65	1	142	end else if (bypass_B_reg) begin
68	1	60	end else if ((red_op_A_reg red_op_B_reg) && (opcode_reg != OR && opcode_reg != XOR)) begin
71	1	752	end else begin
Branch totals: 4 hits of 4 branches = 100.00%			

Total Coverage By Design Unit (filtered view): 99.83%

Instance Coverage													
Instance	Design unit	Design unit type	Total coverage ▾	Stmt count	Stmts hit	Stmts missed	Stmt %	Stmt graph	Branch count	Branches hit	Branches missed	Branch %	Branch graph
/ALU_TB/SVA	ALU_SVA	Module	100.00%	1	1	0	100.00%		2	2	0	100.00%	
/ALU_TB	ALU_TB	Module	100.00%	14	14	0	100.00%						
/ALU_TB/vif	ALU_IF	Module	100.00%										
/ALU_TB/DUT	ALU	Module	99.84%	54	54	0	100.00%		25	25	0	100.00%	

Functional Coverage

Name	Class Type	Coverage	Goal	% of Goal	Status	Included	Merge_instances	Get_inst_coverage	Comment
/ALU_PACKAGE/Coverage		100.00%							
TYPE ALU_Coverage		100.00%	100	100.00...	✓		auto(0)		
INST VALU_PACKAGE::Coverage::ALU_Coverage		100.00%	100	100.00...	✓			0	
CVP ALU_Coverage::RED_OP_B		100.00%	100	100.00...	✓				
CVP ALU_Coverage::RED_OP_A		100.00%	100	100.00...	✓				
CVP ALU_Coverage::OP_CP		100.00%	100	100.00...	✓				
CVP ALU_Coverage::Direction		100.00%	100	100.00...	✓				
CVP ALU_Coverage::BY_PASS_B		100.00%	100	100.00...	✓				
CVP ALU_Coverage::BY_PASS_A		100.00%	100	100.00...	✓				
CVP ALU_Coverage::B_CP		100.00%	100	100.00...	✓				
CVP ALU_Coverage::A_CP		100.00%	100	100.00...	✓				
CROSS ALU_Coverage::SHIFT_LEFT_RIGHT		100.00%	100	100.00...	✓				
CROSS ALU_Coverage::Rotate_LEFT_RIGHT		100.00%	100	100.00...	✓				
CROSS ALU_Coverage::REDUCTION_B		100.00%	100	100.00...	✓				
CROSS ALU_Coverage::REDUCTION_A		100.00%	100	100.00...	✓				
CROSS ALU_Coverage::Multi_OVERFLOW		100.00%	100	100.00...	✓				
CROSS ALU_Coverage::ADDITION_OVERFLOW		100.00%	100	100.00...	✓				

Coverage Report by instance with details

=====

=== Instance: /ALU_PACKAGE

=== Design Unit: work.ALU_PACKAGE

=====

Covergroup Coverage:

Covergroups	1	na	na	100.00%
Coverpoints/Crosses	14	na	na	na
Covergroup Bins	34	34	0	100.00%

Covergroup	Metric	Goal

TYPE /ALU_PACKAGE/Coverage/ALU_Coverage	100.00%	100
covered/total bins:	34	34
missing/total bins:	0	34
% Hit:	100.00%	100
Coverpoint A_CP	100.00%	100
covered/total bins:	4	4
missing/total bins:	0	4
% Hit:	100.00%	100
Coverpoint B_CP	100.00%	100
covered/total bins:	4	4
missing/total bins:	0	4
% Hit:	100.00%	100

Assertion Coverage

Name	Language	Enabled	Log	Count	AtLeast	Limit	Weight	Unmet %	Unmet graph	Included	Memory	Peak Memory	Peak Memory Time	Cumulative Threads
/ALU_TB/SVA/ADDITION_COV	SVA	✓	Off	415	1	Unli...	1	100%		✓	0	0	0 ns	0
/ALU_TB/SVA/MULTIPLICATION_COV	SVA	✓	Off	625	1	Unli...	1	100%		✓	0	0	0 ns	0
/ALU_TB/SVA/ORR_COV	SVA	✓	Off	173	1	Unli...	1	100%		✓	0	0	0 ns	0
/ALU_TB/SVA/REDUCTION_OR_A_COV	SVA	✓	Off	342	1	Unli...	1	100%		✓	0	0	0 ns	0
/ALU_TB/SVA/REDUCTION_OR_B_COV	SVA	✓	Off	163	1	Unli...	1	100%		✓	0	0	0 ns	0
/ALU_TB/SVA/XORR_COV	SVA	✓	Off	94	1	Unli...	1	100%		✓	0	0	0 ns	0
/ALU_TB/SVA/REDUCTION_XOR_A_COV	SVA	✓	Off	247	1	Unli...	1	100%		✓	0	0	0 ns	0
/ALU_TB/SVA/REDUCTION_XOR_B_COV	SVA	✓	Off	138	1	Unli...	1	100%		✓	0	0	0 ns	0
/ALU_TB/SVA/SHIFTING_RIGHT_COV	SVA	✓	Off	153	1	Unli...	1	100%		✓	0	0	0 ns	0
/ALU_TB/SVA/SHIFTING_LEFT_COV	SVA	✓	Off	158	1	Unli...	1	100%		✓	0	0	0 ns	0
/ALU_TB/SVA/Rotate_Right_COV	SVA	✓	Off	121	1	Unli...	1	100%		✓	0	0	0 ns	0
/ALU_TB/SVA/Rotate_Left_COV	SVA	✓	Off	158	1	Unli...	1	100%		✓	0	0	0 ns	0
/ALU_TB/SVA/BY_Pass_A_COV	SVA	✓	Off	260	1	Unli...	1	100%		✓	0	0	0 ns	0
/ALU_TB/SVA/BY_Pass_B_COV	SVA	✓	Off	260	1	Unli...	1	100%		✓	0	0	0 ns	0
/ALU_TB/SVA/INVALID_COV	SVA	✓	Off	45570	1	Unli...	1	100%		✓	0	0	0 ns	0

Coverage Report by instance with details

==== Instance: /ALU_TB/SVA
 === Design Unit: work.ALU_SVA
 =====

Assertion Coverage:
 Assertions

17

17

0

100.00%

Name	Assertion Type	Design Unit	Design UnitType	Lang	File(Line)
/ALU_TB/SVA/ADDITION	Concurrent	ALU_SVA	Verilog	SVA	SVA.sv(182)
/ALU_TB/SVA/MULTIPLICATION	Concurrent	ALU_SVA	Verilog	SVA	SVA.sv(186)
/ALU_TB/SVA/ORR	Concurrent	ALU_SVA	Verilog	SVA	SVA.sv(189)
/ALU_TB/SVA/REDUCTION_OR_A	Concurrent	ALU_SVA	Verilog	SVA	SVA.sv(192)
/ALU_TB/SVA/REDUCTION_OR_B	Concurrent	ALU_SVA	Verilog	SVA	SVA.sv(195)
/ALU_TB/SVA/XORR	Concurrent	ALU_SVA	Verilog	SVA	SVA.sv(198)
/ALU_TB/SVA/REDUCTION_XOR_A	Concurrent	ALU_SVA	Verilog	SVA	SVA.sv(201)

Name	Assertion Type	Language	Enabled	Failure Count	Pass Count	Active Count	Memory	Peak Memory	Peak Memory Time	Cumulative Threads	ATV	Assertion Expression	Included
/ALU_PACKAGE::Generator::run/#ano...	Immediate	SVA	on	0	1	-	-	-	-	-	-	off assert (randomize(...))	✓
/ALU_TB/SVA/ADDITION	Concurrent	SVA	on	0	1	-	0B	0B	0 ns	0	off	assert(@(posedge SVA_JF.clk) dis...	✓
/ALU_TB/SVA/MULTIPLICATION	Concurrent	SVA	on	0	1	-	0B	0B	0 ns	0	off	assert(@(posedge SVA_JF.clk) dis...	✓
/ALU_TB/SVA/ORR	Concurrent	SVA	on	0	1	-	0B	0B	0 ns	0	off	assert(@(posedge SVA_JF.clk) dis...	✓
/ALU_TB/SVA/REDUCTION_OR_A	Concurrent	SVA	on	0	1	-	0B	0B	0 ns	0	off	assert(@(posedge SVA_JF.clk) dis...	✓
/ALU_TB/SVA/REDUCTION_OR_B	Concurrent	SVA	on	0	1	-	0B	0B	0 ns	0	off	assert(@(posedge SVA_JF.clk) dis...	✓
/ALU_TB/SVA/XORR	Concurrent	SVA	on	0	1	-	0B	0B	0 ns	0	off	assert(@(posedge SVA_JF.clk) dis...	✓
/ALU_TB/SVA/REDUCTION_XOR_A	Concurrent	SVA	on	0	1	-	0B	0B	0 ns	0	off	assert(@(posedge SVA_JF.clk) dis...	✓
/ALU_TB/SVA/REDUCTION_XOR_B	Concurrent	SVA	on	0	1	-	0B	0B	0 ns	0	off	assert(@(posedge SVA_JF.clk) dis...	✓
/ALU_TB/SVA/SHIFTING_RIGHT	Concurrent	SVA	on	0	1	-	0B	0B	0 ns	0	off	assert(@(posedge SVA_JF.clk) dis...	✓
/ALU_TB/SVA/SHIFTING_LEFT	Concurrent	SVA	on	0	1	-	0B	0B	0 ns	0	off	assert(@(posedge SVA_JF.clk) dis...	✓
/ALU_TB/SVA/Rotate_Right	Concurrent	SVA	on	0	1	-	0B	0B	0 ns	0	off	assert(@(posedge SVA_JF.clk) dis...	✓
/ALU_TB/SVA/Rotate_Left	Concurrent	SVA	on	0	1	-	0B	0B	0 ns	0	off	assert(@(posedge SVA_JF.clk) dis...	✓
/ALU_TB/SVA/BY_Pass_A	Concurrent	SVA	on	0	1	-	0B	0B	0 ns	0	off	assert(@(posedge SVA_JF.clk) dis...	✓
/ALU_TB/SVA/BY_Pass_B	Concurrent	SVA	on	0	1	-	0B	0B	0 ns	0	off	assert(@(posedge SVA_JF.clk) dis...	✓
/ALU_TB/SVA/INVALID	Concurrent	SVA	on	0	1	-	0B	0B	0 ns	0	off	assert(@(posedge SVA_JF.clk) dis...	✓
/ALU_TB/SVA/Reset_for_Out	Immediate	SVA	on	0	1	-	-	-	-	-	-	assert (SVA_JF.out==0)	✓
/ALU_TB/SVA/Reset_for_LEDS	Immediate	SVA	on	0	1	-	-	-	-	-	-	assert (SVA_JF.leds==0)	✓

Conclusion

This project successfully demonstrates the design and implementation of a complete **UVM-Inspired Class-Based Verification Environment** for a pipelined Arithmetic Logic Shift Unit (ALU) using System Verilog. Through a structured and reusable verification architecture, all major ALU functionalities — including arithmetic, logical, shift, rotate, reduction, bypass, and invalid operations — were thoroughly verified.

The environment achieved **100% code coverage, high functional coverage** with effective cross-coverage, and strong assertion coverage. Key challenges such as 2-cycle pipeline latency synchronization, virtual interface binding, scoreboard alignment, and coverage closure were systematically identified and resolved, showcasing strong debugging and verification skills.

This project highlights proficiency in advanced verification methodologies, including:

- Constrained-random stimulus generation
- Transaction-level modeling using mailboxes
- Self-checking scoreboards with golden reference models
- Functional and cross-coverage-driven verification
- Assertion-based verification (SVA)
- Pipeline-aware testbench design

The successful completion of this project with full verification closure reflects a solid understanding of modern ASIC verification flows and best practices. The developed environment serves as a reusable framework for future verification projects and demonstrates readiness for real-world digital verification roles.